

Car Price Prediction Machine Learning Web Application

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Project Overview

The Car Price Prediction Machine Learning Web Application is an end-to-end machine learning project designed to estimate the selling price of used cars based on various vehicle characteristics. The project combines data preprocessing, machine learning model development, web application deployment, and cloud hosting to provide a user-friendly prediction system.

Users can enter vehicle information such as brand, manufacturing year, fuel type, transmission type, mileage, engine capacity, and other specifications to receive an estimated car price instantly.

Objectives

- Build a machine learning model capable of predicting used car prices.
 - Develop an interactive web interface for real-time predictions.
 - Deploy the application online for public access.
 - Gain practical experience in the complete machine learning lifecycle.
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Technologies Used

Programming Language

- Python

Data Analysis & Processing

- Pandas
- NumPy

Machine Learning

- Scikit-Learn
- Linear Regression

Model Serialization

- Pickle

Web Application

- Streamlit

Version Control & Deployment

- Git
 - GitHub
 - Streamlit Community Cloud
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Dataset

Dataset Name

Used Car Dataset (Cardetails.csv)

Features Used

- Car Brand
- Manufacturing Year
- Kilometers Driven
- Fuel Type
- Seller Type
- Transmission Type

- Ownership Status
- Mileage
- Engine Capacity (CC)
- Maximum Power
- Number of Seats

Target Variable

- Selling Price
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Data Preprocessing

Several preprocessing steps were performed before model training:

Data Cleaning

- Removed unnecessary data.
- Handled missing values.
- Checked for duplicate records.

Feature Engineering

- Extracted vehicle brand names from car titles.
- Selected relevant attributes for training.

Categorical Encoding

Converted categorical values into numerical values:

Examples:

Feature	Encoding
Fuel Type	Diesel → 1, Petrol → 2, LPG → 3, CNG → 4
Transmission	Manual → 1, Automatic → 2
Seller Type	Individual → 1, Dealer → 2, Trustmark Dealer → 3

Data Type Conversion

Converted model input values into numeric format before prediction to ensure compatibility with Scikit-Learn models.

Machine Learning Model

Algorithm Used

Linear Regression

Training Process

1. Dataset loading
2. Data preprocessing
3. Feature selection
4. Feature encoding
5. Train-test split
6. Model training
7. Performance evaluation
8. Model serialization using Pickle

Model Saving

```
import pickle

pickle.dump(model, open("model.pkl", "wb"))
```

The trained model was stored as:

```
model.pkl
```

Streamlit Web Application

Features

- Interactive user interface
- Dropdown selection for categorical features
- Sliders for numerical features
- Real-time price prediction
- Simple and responsive design

User Inputs

Users provide:

- Car Brand
- Manufacturing Year
- Fuel Type
- Seller Type
- Transmission Type
- Ownership Status
- Mileage
- Engine Capacity
- Maximum Power
- Number of Seats
- Kilometers Driven

Output

The application predicts and displays:

Estimated Car Price

Deployment Process

GitHub Repository

The project files were uploaded to GitHub:

```
app.py  
model.pkl  
Cardetails.csv  
requirements.txt  
Car_Price_Prediction.ipynb
```

Streamlit Cloud Deployment

Deployment steps:

1. Push project to GitHub.
2. Connect GitHub repository to Streamlit Cloud.
3. Configure deployment settings.
4. Deploy application.
5. Generate public URL.

The application became accessible through a live web link.

Challenges Faced During Deployment

1. Missing Model File Issue

Problem:

```
FileNotFoundError
```

Cause:

- Streamlit Cloud could not locate the model file.

Solution:

- Used proper file path handling with `pathlib`.
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2. Missing Dependencies

Problem:

```
ModuleNotFoundError: No module named 'sklearn'
```

Cause:

- Required libraries were not installed in the deployment environment.

Solution:

Created a proper requirements file:

```
streamlit  
pandas  
numpy  
scikit-learn
```

3. Prediction Data Type Error

Problem:

```
ValueError during model.predict()
```

Cause:

- Some input values remained as string/object data types.

Solution:

```
input_data_model = input_data_model.apply(pd.to_numeric)
```

All model inputs were converted to numeric format before prediction.

Skills Demonstrated

Machine Learning

- Data preprocessing
- Feature engineering
- Model training
- Model evaluation
- Model serialization

Software Development

- Python programming
- Streamlit development
- Error debugging
- Dependency management

Deployment

- Git version control
 - GitHub repository management
 - Cloud deployment
 - Production troubleshooting
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Key Learning Outcomes

Through this project, I gained practical experience in:

- Building end-to-end machine learning solutions
 - Developing interactive web applications
 - Deploying ML applications to cloud platforms
 - Managing project dependencies
 - Debugging deployment issues
 - Converting research notebooks into production-ready applications
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Resume Description

- Developed a Car Price Prediction Machine Learning application using Python, Scikit-Learn, and Streamlit.
 - Performed data preprocessing, feature engineering, and model training on a real-world used car dataset.
 - Built and deployed an interactive web application on Streamlit Cloud for real-time vehicle price estimation.
 - Utilized Git, GitHub, and Pickle for version control, deployment, and model serialization.
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LinkedIn Project Description

Car Price Prediction Machine Learning Web Application

Built an end-to-end machine learning application that predicts used car prices based on vehicle specifications. The project involved data preprocessing, feature engineering, model training using Scikit-Learn, and deployment through Streamlit Cloud.

Technologies Used

Python | Pandas | NumPy | Scikit-Learn | Streamlit | Git | GitHub

Key Highlights

- ✓ Data Cleaning & Preprocessing
 - ✓ Feature Engineering
 - ✓ Machine Learning Model Development
 - ✓ Streamlit Web Application
 - ✓ Cloud Deployment
 - ✓ Real-Time Car Price Prediction
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Interview Explanation

"This project is an end-to-end machine learning application developed to predict used car prices. I started by collecting and preprocessing a real-world car dataset, handling categorical variables through encoding techniques and preparing the data for model training. I trained a Linear Regression model using Scikit-Learn and saved the trained model using Pickle. After training, I developed an interactive web application using Streamlit where users can input vehicle information and receive a predicted car price instantly. Finally, I deployed the application on Streamlit Cloud using GitHub. During deployment, I solved issues related to dependency management, model loading, file paths, and data type mismatches, which helped me gain valuable experience in taking machine learning projects from development to production."